

Exercise: Numerical Integration of a Radiating Binary and Gravitational Waveform

Overview

In this exercise you will integrate the trajectory of a two-body system under Newtonian gravity plus the 2.5PN radiation reaction acceleration. You will then verify that the orbital decay follows the Peters equations, and finally compute the gravitational waveform using the quadrupole formula. Throughout, keep track of both bodies separately.

Part 1: Equations of Motion

The equations of motion for body 1 including Newtonian gravity and the 2.5PN radiation reaction term are (from here, eq. 339):

$$\begin{aligned} \mathbf{a}_1 = & -\frac{Gm_2}{r_{12}^2} \mathbf{n}_{12} \\ & + \frac{1}{c^5} \left\{ \left[\frac{208G^3m_1m_2^2}{15r_{12}^4} (n_{12}v_{12}) - \frac{24G^3m_1^2m_2}{5r_{12}^4} (n_{12}v_{12}) + \frac{12G^2m_1m_2}{5r_{12}^3} (n_{12}v_{12})v_{12}^2 \right] \mathbf{n}_{12} \right. \\ & \left. + \left[\frac{8G^3m_1^2m_2}{5r_{12}^4} - \frac{32G^3m_1m_2^2}{5r_{12}^4} - \frac{4G^2m_1m_2}{5r_{12}^3} v_{12}^2 \right] \mathbf{v}_{12} \right\} \end{aligned} \quad (1)$$

The equations for body 2 are obtained by the exchange $1 \leftrightarrow 2$, remembering that $\mathbf{n}_{12}$ and $\mathbf{v}_{12}$ change sign under this operation ($\mathbf{n}_{21} = -\mathbf{n}_{12}$).

1. Consider adequate internal units, e.g. when $G = c = M = 1$.
2. Reduce both equations of motion to a system of first-order ODEs for the state vector $(\mathbf{r}_1, \mathbf{r}_2, \mathbf{v}_1, \mathbf{v}_2)$ and integrate them numerically using a method of your choice (e.g. Runge-Kutta 4). Choose initial conditions corresponding to a quasi-circular orbit with a given semi-major axis a_0 and eccentricity e_0 .
3. At each time step, compute the instantaneous orbital elements $a(t)$ and $e(t)$ from the relative coordinate $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$ and relative velocity $\mathbf{v} = \mathbf{v}_1 - \mathbf{v}_2$, using:

$$\mathcal{E} = \frac{v^2}{2} - \frac{GM}{r}, \quad a = -\frac{GM}{2\mathcal{E}}, \quad (2)$$

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}, \quad e = \sqrt{1 - \frac{h^2}{GMa}}, \quad (3)$$

where $M = m_1 + m_2$, $r = |\mathbf{r}|$, $v = |\mathbf{v}|$, $h = |\mathbf{h}|$.

4. Be aware that you cannot integrate down to the merger, so you need to set a max time (i.e. the time to merger, see below), or enforce a check on the value of the separation vector (i.e. $r > 6$, using appropriate units).

Part 2: Comparison with the Peters Equations

The orbit-averaged evolution of the semi-major axis and eccentricity under gravitational wave emission is given by the Peters (1964) equations:

$$\left\langle \frac{da}{dt} \right\rangle = -\frac{64}{5} \frac{G^3 m_1 m_2 M}{c^5 a^3} f(e), \quad (4)$$

$$\left\langle \frac{de}{dt} \right\rangle = -\frac{304}{15} e \frac{G^3 m_1 m_2 M}{c^5 a^4} g(e), \quad (5)$$

where the enhancement functions due to eccentricity are:

$$f(e) = \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{(1 - e^2)^{7/2}}, \quad (6)$$

$$g(e) = \frac{1 + \frac{121}{304}e^2}{(1 - e^2)^{5/2}}. \quad (7)$$

Note that $f(0) = g(0) = 1$ for circular orbits.

1. Integrate equations (4)–(5) numerically as a coupled system for $a(t)$ and $e(t)$, starting from the same initial conditions a_0 and e_0 as in Part 1.
2. Compare the evolution of $a(t)$ and $e(t)$ obtained from the full 2-body integration (Part 1) with the Peters prediction. Plot both on the same axes and comment on the agreement.
3. Optional: Estimate the merger timescale T_{merger} from both methods, defined as the time at which $a \rightarrow 0$. For a circular orbit, an analytic estimate is:

$$T_{\text{merger}} = \frac{a_0^4}{4\beta} \quad (8)$$

while for eccentric orbits you need to compute this integral numerically

$$T_{\text{merger}} = \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{29/19} (1 + \frac{121}{304}e^2)^{1181/2299}}{(1 - e^2)^{3/2}} de, \quad (9)$$

$$\text{where } \beta = \frac{64}{5} \frac{G^3 m_1 m_2 M}{c^5} \text{ and } c_0 = \frac{a_0(1 - e_0^2)}{e_0^{12/19}} \left(1 + \frac{121}{304}e_0^2\right)^{-870/2299}.$$

Part 3: Gravitational Waveform from the Quadrupole Formula

The gravitational wave strain in the transverse-traceless (TT) gauge at distance R from the source is:

$$h_{\text{TT}}^{ij}(t, \mathbf{x}) = \frac{2G}{c^4 R} \Lambda^{ij}_{kl}(\mathbf{n}) \ddot{Q}^{kl}(t_{\text{ret}}), \quad (10)$$

where $t_{\text{ret}} = t - R/c$ is the retarded time, $\mathbf{n} = \mathbf{x}/R$ is the unit vector pointing from the source to the observer, and $\Lambda^{ij}_{kl}(\mathbf{n})$ is the **TT projection operator**, defined as:

$$\Lambda^{ij}_{kl}(\mathbf{n}) = P^i_k P^j_l - \frac{1}{2} P^{ij} P_{kl}, \quad P^{ij} = \delta^{ij} - n^i n^j. \quad (11)$$

Here P^{ij} is the transverse projector onto the plane orthogonal to $\mathbf{n}$, and Λ^{ij}_{kl} further removes the trace. The TT operator satisfies $\Lambda^{ij}_{kl} n^k = 0$ (transversality) and $\Lambda^{ij}_{ij} = 0$ (tracelessness).

The **trace-free** (STF) mass quadrupole moment is:

$$Q^{ij} = I^{ij} - \frac{1}{3} \delta^{ij} I^{kk}, \quad (12)$$

with the second mass moment for the two-body system given explicitly by:

$$I^{ij}(t) = m_1 r_1^i(t) r_1^j(t) + m_2 r_2^i(t) r_2^j(t). \quad (13)$$

Its first and second time derivatives, which you can compute from the stored trajectory, are:

$$\dot{I}^{ij} = m_1 \left(v_1^i r_1^j + r_1^i v_1^j \right) + m_2 \left(v_2^i r_2^j + r_2^i v_2^j \right), \quad (14)$$

$$\ddot{I}^{ij} = m_1 \left(a_1^i r_1^j + 2v_1^i v_1^j + r_1^i a_1^j \right) + m_2 \left(a_2^i r_2^j + 2v_2^i v_2^j + r_2^i a_2^j \right), \quad (15)$$

where $\mathbf{v}_A = \dot{\mathbf{r}}_A$ and $\mathbf{a}_A = \ddot{\mathbf{r}}_A$ with $A = 1, 2$ are directly available from the numerical integration. Note that since Q^{ij} and I^{ij} differ only by a trace term proportional to δ^{ij} , and the TT projector removes the trace, one has $\Lambda^{ij}_{kl} \ddot{Q}^{kl} = \Lambda^{ij}_{kl} \ddot{I}^{kl}$ in practice.

Extracting the Two Polarisations from h_{ij}^{TT}

Degrees of freedom of h_{ij}^{TT} . The TT strain is a 3×3 matrix subject to three conditions:

- symmetry: $h_{ij} = h_{ji}$,
- transversality: $h_{ij}n^j = 0$,
- tracelessness: $h^i_i = 0$.

These conditions together reduce the number of independent real degrees of freedom to exactly **two**, which are the two polarisation amplitudes h_+ and $h_\times$.

Polarisation tensor basis. To extract these two numbers from the matrix h_{ij}^{TT} , introduce two unit vectors $\mathbf{e}_1$ and $\mathbf{e}_2$, both orthogonal to $\mathbf{n}$ and to each other:

$$\mathbf{e}_1 \cdot \mathbf{n} = \mathbf{e}_2 \cdot \mathbf{n} = 0, \quad \mathbf{e}_1 \cdot \mathbf{e}_2 = 0, \quad \mathbf{e}_1 \cdot \mathbf{e}_1 = \mathbf{e}_2 \cdot \mathbf{e}_2 = 1. \quad (16)$$

Define the two polarisation tensors:

$$e_{ij}^+ = e_{1i}e_{1j} - e_{2i}e_{2j}, \quad e_{ij}^\times = e_{1i}e_{2j} + e_{2i}e_{1j}. \quad (17)$$

These form a complete, orthogonal basis for the space of symmetric, transverse, traceless tensors with respect to $\mathbf{n}$, and satisfy the orthonormality relations:

$$e^{+ij}e_{ij}^+ = 2, \quad e^{\times ij}e_{ij}^\times = 2, \quad e^{+ij}e_{ij}^\times = 0. \quad (18)$$

Decomposition of h_{ij}^{TT} . Since e_{ij}^+ and $e_{ij}^\times$ span the TT space, we can always write:

$$h_{ij}^{\text{TT}} = h_+ e_{ij}^+ + h_\times e_{ij}^\times. \quad (19)$$

To extract h_+ and $h_\times$, contract both sides with the respective polarisation tensor and use the orthonormality relations above:

$$h_+ = \frac{1}{2} e^{+ij} h_{ij}^{\text{TT}}, \quad h_\times = \frac{1}{2} e^{\times ij} h_{ij}^{\text{TT}}, \quad (20)$$

where the factor $1/2$ comes from the normalisation $e^{+ij}e_{ij}^+ = 2$.

Substituting the quadrupole formula. Substituting $h_{ij}^{\text{TT}} = (2G/c^4 R) \Lambda_{ij}{}^{kl} \ddot{Q}_{kl}$ and noting that e_{ij}^+ and $e_{ij}^\times$ are themselves transverse and traceless with respect to $\mathbf{n}$, the contraction with the TT projector is trivial:

$$e^{+ij} \Lambda_{ij}{}^{kl} = e^{+kl}, \quad e^{\times ij} \Lambda_{ij}{}^{kl} = e^{\times kl}. \quad (21)$$

Therefore the $1/2$ from the normalisation cancels the factor of 2 in the quadrupole formula, and one obtains:

$$h_+ = \frac{G}{c^4 R} e^{+ij} \ddot{Q}_{ij} = \frac{G}{c^4 R} (e_1^i e_1^j - e_2^i e_2^j) \ddot{Q}_{ij}, \quad (22)$$

$$h_\times = \frac{G}{c^4 R} e^{\times ij} \ddot{Q}_{ij} = \frac{2G}{c^4 R} e_1^i e_2^j \ddot{Q}_{ij}, \quad (23)$$

where the last step used the symmetry of $\ddot{Q}_{ij}$.

Face-on and edge-on cases. For $\mathbf{n} = \mathbf{e}_z$, choose $\mathbf{e}_1 = \mathbf{e}_x$ and $\mathbf{e}_2 = \mathbf{e}_y$. Then $e_{ij}^+ = \text{diag}(1, -1, 0)$ and $e_{ij}^\times$ has only off-diagonal entries $e_{xy}^\times = e_{yx}^\times = 1$. Plugging into equations (22)–(23):

$$h_+ = \frac{G}{c^4 R} (\ddot{Q}^{xx} - \ddot{Q}^{yy}), \quad h_\times = \frac{2G}{c^4 R} \ddot{Q}^{xy}, \quad (24)$$

which is the standard result for a face-on binary.

Repeat for $\mathbf{n} = \mathbf{e}_x$, choose $\mathbf{e}_1 = -\mathbf{e}_z$ and $\mathbf{e}_2 = \mathbf{e}_y$ for the edge-on case.

Hint: you might want to express the $\mathbf{e}$ vectors in terms of spherical coordinates, to easily change the orientation.

Visualise. Plot $h_+(t)$ and $h_\times(t)$ for both orientations and verify:

- the frequency of the waveform is twice the orbital frequency;
- the magnitude of polarisation changes with different line of sight;
- compare with literature results (e.g. chapter 4 of Maggiore vol1).

Part 4 (Optional): Verification of energy balance

Verify energy conservation by computing the GW luminosity:

$$L_{\text{GW}} = \frac{G}{5c^5} \ddot{Q}^{ij} \ddot{Q}^{ij}, \quad (25)$$

and checking that $L_{\text{GW}} \approx -\dot{\mathcal{E}}_{\text{orb}}$, where $\dot{\mathcal{E}}_{\text{orb}}$ is the time derivative of the orbital energy. This requires evaluating $\ddot{Q}^{ij}$: think about how to compute a numerical third derivative from the methods you have studied. Which finite-difference scheme would you use, and how does the choice of step size affect the accuracy? Compare the result with the analytic Peters luminosity and comment on any discrepancy, if any.